Lab Assignment 04

Unsharp Masking and High-Boost Filtering

Write a program that takes a grayscale image as user input and performs:

1. **Unsharp masking**
2. **High-boost filtering**

The program should allow the user to select the required parameters.

Required Steps

First, apply a smoothing filter to the input image:

Blurred Image = Low-pass Filter(Image)

Then calculate the mask:

Mask = Original Image − Blurred Image

01. Unsharp Masking

Generate the sharpened image using:
Sharpened Image = Original Image + Mask
or equivalently:

$$g(x,y) = f(x,y) + [f(x,y) - \bar{f}(x,y)]$$

The program should display:

1. Original image
2. Blurred image
3. Mask
4. Unsharp-masked image

02. High-Boost Filtering

Implement high-boost filtering using:

$$g(x,y) = A\, f(x,y) - \bar{f}(x,y)$$

Where: $A > 1$

The value of must be provided by the user.
Test the algorithm using at least three different values, such as:

$A = 1.5$
$A = 2.0$
$A = 3.0$

Display the results and compare the effect of different values.